Creation of a Machine Learning Model and Mobile Application for the Detection of Heart Murmurs

Introduction

- Heart murmurs are abnormal heart sounds (like whooshing)
- Heart murmurs can serve as indicators for more serious conditions, like aneurysms
- Access to heart screenings and similar medical care is expensive and hard to find in many areas
- Creation of a mobile application that classifies heart sounds it records as potentially having murmurs

Methods

- Train machine learning models with recordings from four heart valves
- Utilize React Native Executorch to export the model and run it on a mobile device
- Record and process audio within the app created with React Native
- Oversampling via SMOTE to account for imbalance in dataset
- Compared traditional methods with hyperparameter optimization

Results

- 0.7-0.9 accuracy in machine learning training using traditional methods (varies by location)
- 0.7-0.8 F1 score
- Comparatively higher than 0.2-0.3 F1 score with hyperparameter optimization and without oversampling via SMOTE
- App not fully complete
  - Audio processing not working, otherwise complete

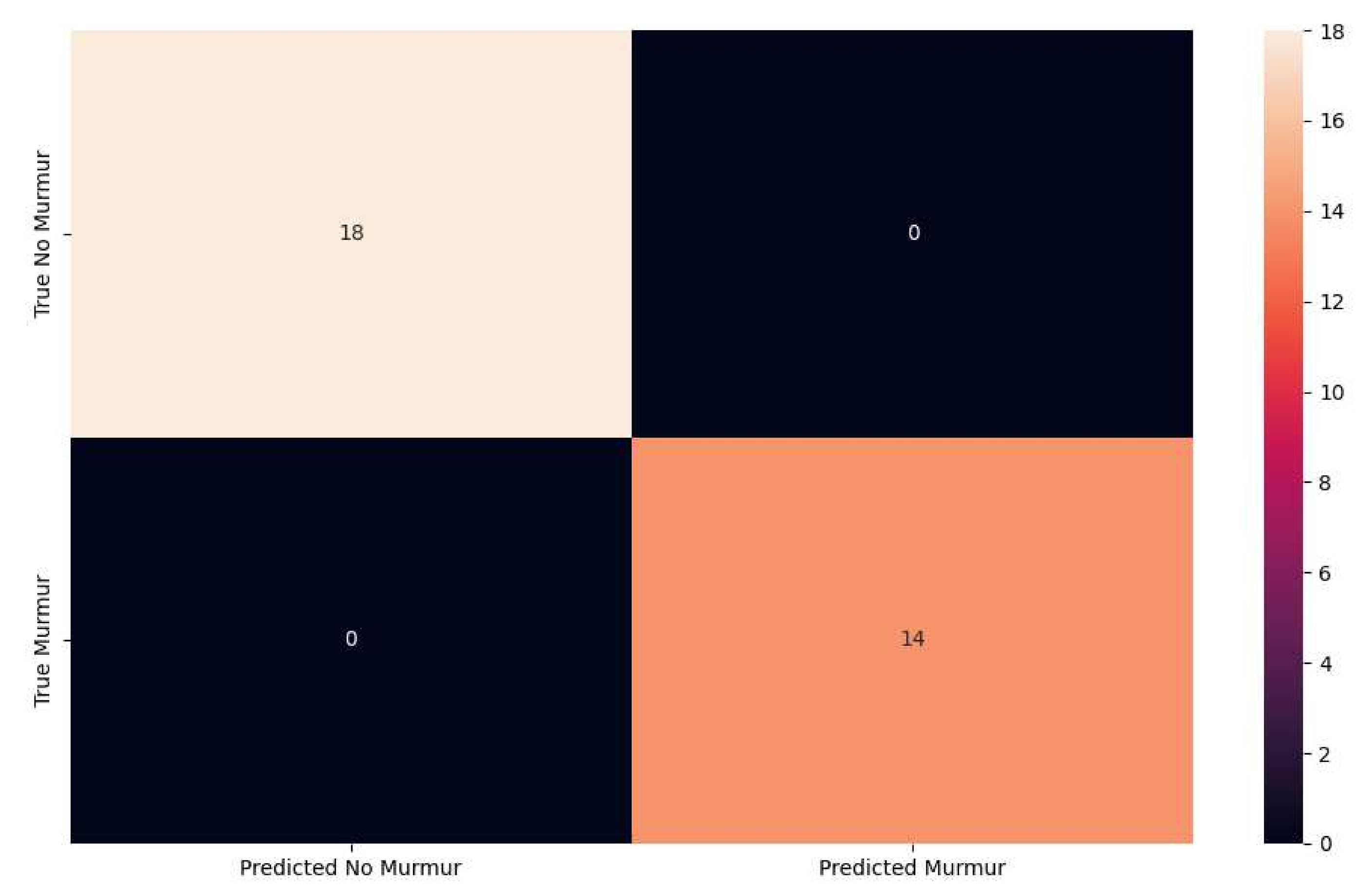

Figure 1: Confusion matrix from training for one of the machine learning models

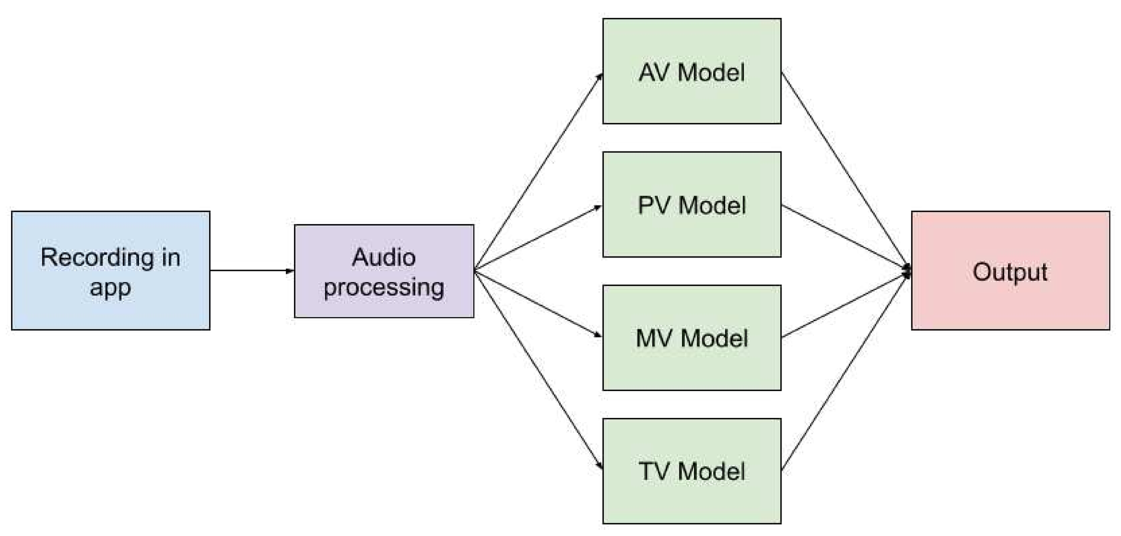

Figure 2: Overall Classification Process Diagram

Conclusions

- Hyperparameter optimization less effective than normal training for heart murmur audio
- Heart murmurs can be classified with machine learning
- Ongoing issue with audio processing and isolating the murmur sound
- There is definitely work that still needs to be done

References

- Baheti, P. (2023, July 27). Working with audio data for machine learning in Python. Comet. https://www.comet.com/site/blog/working-with-audio-data-for-machine-learning-in-python/
- Heart murmur: Symptoms & treatments. University of Utah Health. (2025, January 23). https://healthcare.utah.edu/cardiovascular/conditions/heart-murmur
- Kang, S. H., Joe, B., Yoon, Y., Cho, G. Y., Shin, I., & Suh, J. W. (2018). Cardiac Auscultation Using Smartphones: Pilot Study. JMIR mHealth and uHealth, 6(2), e49. https://doi.org/10.2196/mhealth.8946
- Kent, K. C. (2014). Abdominal aortic aneurysms. New England Journal of Medicine, 371(22), 2101–2108. https://doi.org/10.1056/nejmcp1401430
- Levin, A. D., Ragazzi, A., Szot, S. L., & Ning, T. (2020). A machine learning approach to heart murmur detection and classification. 2020 13th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI), 521–525. https://doi.org/10.1109/cisp-bmei51763.2020.9263528
- Oliveira, J., Renna, F., Costa, P., Nogueira, M., Oliveira, A. C., Elola, A., Ferreira, C., Jorge, A., Bahrami Rad, A., Reyna, M., Sameni, R., Clifford, G., & Coimbra, M. (2022). The CirCor DigiScope Phonocardiogram Dataset (version 1.0.3). PhysioNet. RRID:SCR_007345. https://doi.org/10.13026/tshs-mw03
- Potdar, R. M., Meshram, M. R., & Kumar, R. (2021). Cardiovascular Condition Monitoring based on Heart Murmur Analysis. Turkish Online Journal of Qualitative Inquiry, 12(6), 156–166.
- Rios-Gutierrez, F., Alba-Flores, R., Ejaz, K., Nordehn, G., Andrisevic, N., & Burns, S. (2006). Classification of four types of common murmurs using wavelets and a learning vector quantization network. The 2006 IEEE International Joint Conference on Neural Network Proceedings, 2206–2213. https://doi.org/10.1109/ijcnn.2006.1716385